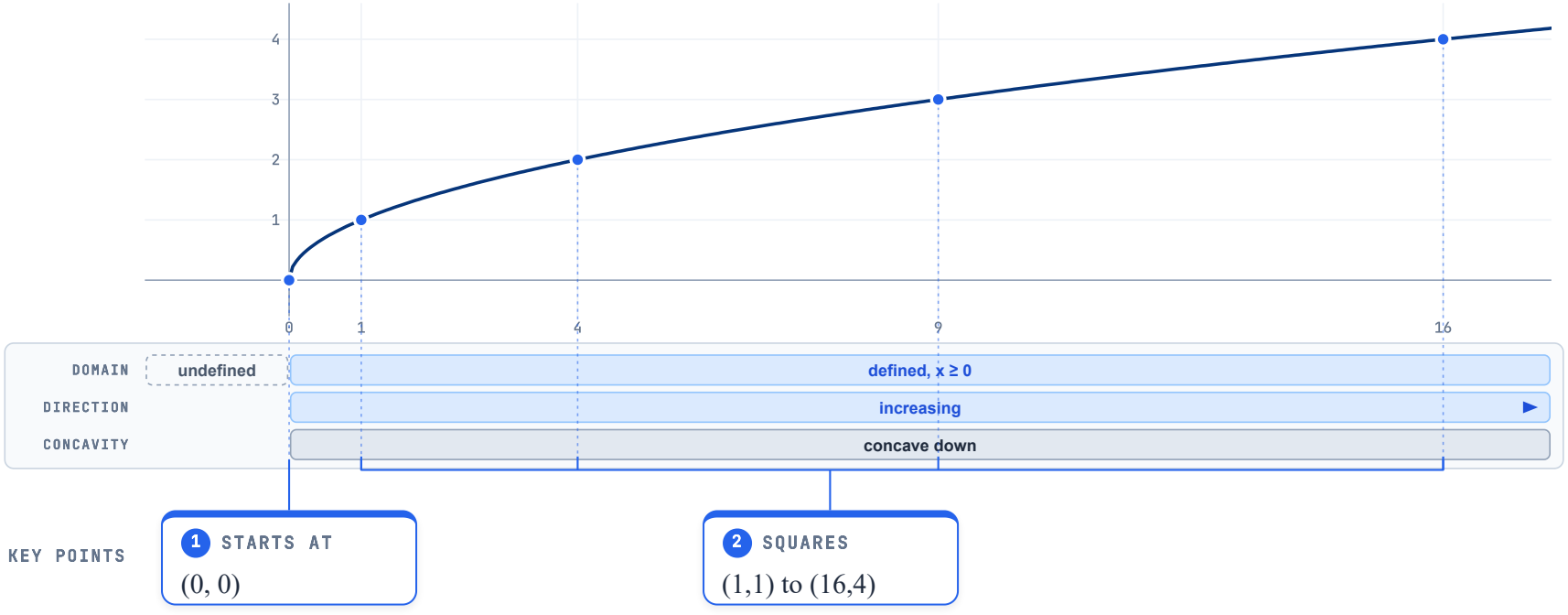
PARENT FUNCTION

Profile of the square root

$f(x) = \sqrt{x}$

[0, ∞) DOMAIN

1-TO-1 INVERTIBLE



KEY POINTS

1 STARTS AT

(0, 0)

2 SQUARES

(1,1) to (16,4)

GLOBAL

DOMAIN

$[0, \infty)$

only non-negative inputs

RANGE

$[0, \infty)$

principal root is never negative

DIRECTION

increasing throughout

SHAPE

concave down

the rise slows

ONE-TO-ONE

yes

passes the horizontal line test